

A review of active cell balancing methods in electric vehicles

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ABSTRACT

The high energy density and modularity of Lithium-ion (Li-ion) batteries make them ideal for electric vehicles (EVs). However, in series-connected battery packs, cell imbalance emerges as a critical challenge, resulting in voltage variations, accelerated aging, safety concerns, and reduced usable capacity. This paper systematically compares and evaluates active cell balancing methods, presenting a comparative analysis to provide a suitable selection for EVs. Furthermore, it offers a classification of DC-DC converter-based balancing circuits, distinguishing between non-isolated and isolated configurations based on their electrical isolation characteristics, analyzing their trade-offs in efficiency, cost, and balancing speed. To validate these topologies, four DC-DC converter-based balancing structures are modeled and simulated in MATLAB/Simulink. The simulation results provide a comparative performance analysis, with particular emphasis on balancing speed and current ripple characteristics. Based on these findings, the paper discusses the relative characteristics of the four methods. Finally, potential research directions for optimizing cell balancing techniques are outlined to guide future studies in this field.

1. Introduction

With the growing urgency of the global energy crisis and the accelerating transition toward decarbonization, the demand for energy technologies that are both highly efficient and environmentally sustainable is steadily rising [1]. Lithium-ion batteries, characterized by their high energy density and long cycle lifespan, have found widespread application in EVs [2–4]. To meet the energy requirements of EVs, large-scale battery packs are assembled by connecting cells in series and parallel to provide sufficient voltage and capacity. However, in practical applications, inconsistencies in cell manufacturing and degradation over time result in differences in internal resistance, capacity, and voltage. These disparities accumulate with each charge–discharge cycle, reducing pack efficiency, lowering usable capacity, and accelerating aging. In severe cases, such imbalances can lead to an explosion and even fire [5,6]. To address these challenges, a control and monitoring unit known as the battery management system (BMS) is essential for ensuring that each cell within the battery pack operates within optimal conditions. Among its core functions, cell balancing is a primary focus of BMS design, and has therefore become a key topic in ongoing research.

According to the energy redistribution mechanism, cell balancing circuits (CBC) are generally categorized into passive (dissipative) and active (non-dissipative) types [7,8]. Passive balancing equalizes the state of charge (SoC) by dissipating excess energy from higher-charged cells through shunt resistors in the form of heat. While passive methods are cost-effective and straightforward to implement, they are limited by low energy efficiency, excessive heat generation, and extended balancing durations. The balancing current must be limited to mitigate power dissipation and thermal challenges, inevitably prolonging the balancing duration [9]. Active balancing achieves charge balance through controlled energy transfer instead of dissipation, thereby enhancing energy utilization efficiency [10]. The process generally involves temporarily storing energy extracted from overcharged cells using inductors, capacitors, or converters, and subsequently delivering it to undercharged cells to achieve a balanced state. Owing to their ability to minimize energy losses and accelerate the balancing process, active cell balancing methods are markedly more efficient than passive techniques. These performance benefits have made them an increasingly attractive subject of investigation in the field of BMS.

Numerous studies have been conducted to evaluate and compare the performance of various active cell balancing techniques. In [11], these

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strategies are systematically categorized according to their energy storage components. Building upon this foundation, the present work delivers a detailed overview and evaluation of multiple active balancing topologies. A comparative table (Table 1. Classification of active balancing methods) is also provided to support the selection of suitable methods for different practical applications. Distinguishing itself from conventional research that primarily emphasizes hardware implementations or specific balancing pathways, the study in [12] introduces an innovative and structured classification of active balancing techniques. This framework systematically differentiates between adjacent, non-adjacent, direct cell-to-cell, and mixed configurations, thereby enabling a more comprehensive and nuanced evaluation of various strategies.

Among these strategies, DC-DC converter-based balancing circuits have been the focus of extensive research due to their inherent advantages in terms of compact structure, cost-effectiveness, high balancing efficiency, and fast energy redistribution, when compared with capacitor-based or transformer-based counterparts. Ref [6] provides an in-depth analysis of these circuits, including their control strategies and SoC estimation techniques, both of which are crucial for achieving optimal balancing performance. Refs [13] and [14] both investigate and review bidirectional DC-DC converters, addressing topics ranging from general classification to specific topological structures. Ref [13] emphasizes classification and application-oriented comparisons of various topologies, offering a reference framework for selecting or designing converters. In contrast, Ref [14] focuses on the operational aspects, summarizing widely adopted control and switching strategies across different converter types. Nevertheless, the majority of current research is limited to theoretical-level analysis and comparison of balancing hardware circuits, without experimental or simulation-supported systematic evaluation. To rectify this deficiency, Ref [15] performs a structured extraction and evaluation of recently proposed DC-DC converter-based balancing circuits via simulation, but its assessment is limited to a comparison of balancing speeds.

Several studies have investigated active cell balancing techniques for EVs; however, none have systematically compared balancing speed and current ripple through simulation. This work addresses that gap by making the following contributions:

- A comparative review of active balancing strategies for EV batteries.
- A structured classification of active balancing methods based on energy transfer direction and circuit components.
- A categorization of DC-DC converter-based balancing circuits into non-isolated and isolated configurations, including analysis of trade-offs in efficiency, cost, and balancing speed.

- A simulation-based evaluation of balancing speed and current ripple for four representative DC-DC converter topologies using MATLAB/Simulink.
- Recommendations for future research to optimize balancing topology design in EVs.

This paper provides a focused review of active balancing techniques due to their superior efficiency and adaptability. A survey of recent literature evaluates both the performance and applicability of these strategies. Section 2 introduces two classification schemes: one based on energy transfer direction and the other on energy transfer components. Section 3 examines DC-DC converter-based balancing methods, particularly relevant for EV applications, and distinguishes between non-isolated and isolated converter topologies. Section 3.3 includes a summary table outlining the key characteristics of each topology. Section 4 presents a comparative simulation study assessing the balancing speed and output current ripple of four selected converters. Finally, Section 5 highlights future research directions, and Section 6 provides the conclusions.

2. Active cell balancing methods

Based on whether energy is dissipated or redistributed, cell balancing methods fall into two primary categories: passive and active. In passive balancing, excess energy from cells with elevated charge levels is dissipated as heat using resistive elements. Although it is a cost-effective and easily implementable solution, its inherent inefficiency and substantial heat generation present significant challenges in thermal management and overall system efficiency [16]. The active balancing method functions by transferring surplus energy from an overcharged cell to an undercharged one via an additional balancing circuit, thereby enhancing charge uniformity. Based on the energy transfer mechanism and the components utilized, active balancing circuits are classified into four distinct categories: capacitor-based, inductor-based, transformer-based and DC-DC converter-based [15]. The direction of energy flow represents another key consideration in balancing topology design, requiring each circuit to function with an appropriate energy transfer configuration. A classification of active cell balancing methods is illustrated in Fig. 1.

2.1. Classification based on the direction of energy flow

Active balancing supports multiple energy transfer modes, such as cell bypass, adjacent cell-to-cell (ACTC), direct cell-to-cell (DCTC), cell-to-pack (CTP), pack-to-cell (PTC), cell-to-pack-to-cell (CTPTC) and multicell-to-multicell (MCTMC), allowing for flexible and efficient

Table 1
Comparison of Different Battery Balancing Topologies [18–32].

Topology	Speed	Control Complexity	Efficiency	Advantage	Disadvantage	Application
Cell Bypass	High	Medium	Low	No bulky converters, low cost, small size, high speed	Voltage drop with many bypassed cells, low efficiency, used for low power application, high switch current stress	System with low power and moderate balancing
ACTC	Low	Low	Medium	Simple control, scalable, efficient charge transfer mechanism	Inefficient for distant cell pairs, energy loss over multiple transfer	Small scale battery pack
DCTC	Medium	Medium	High	Fast balancing, compact design using shared converter	High switch count, only two cells active per cycle	High-voltage and compact battery system
CTP	Medium	Medium	Low	Faster balancing due to large voltage difference, simple circuit architecture, less cost	Low efficiency due to multiple conversion stages	Cost-sensitive large-scale packs
PTC	Medium	Medium	Low	Effective in recharging deeply discharged cells, centralized control	Transformer losses, low efficiency	Centralized BMS with deep-discharge recovery
CTPTC	High	High	Medium	Combines advantages of CTP and PTC, high speed, flexible energy routing	Bulky, expensive, high control complexity	Systems requiring dynamic and flexible energy routing
MCTMC	High	High	High	Parallel balancing, high speed and efficiency, flexible operation modes	High control complexity, advanced control and switching matrix	High-performance, large-scale systems

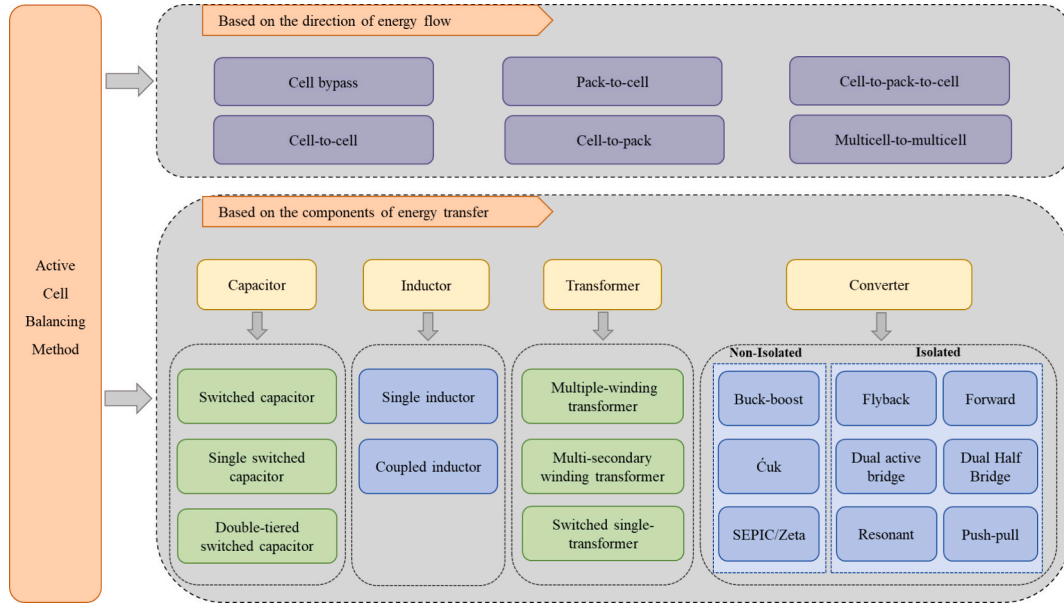


Fig. 1. Active cell balancing methods.

charge redistribution. A detailed analysis of their respective merits and constraints is presented in this subsection.

2.1.1. Cell bypass

Cell bypass balancing is a battery management strategy designed to redistribute current by diverting the flow from cells that have reached their upper voltage or SoC limit until uniform state-of-charge levels are achieved across all cells. This technique is classified into three primary categories: complete shunting, shunt resistor, and shunt transistor methods [17]. Unlike traditional configurations, where current flows uniformly through all cells, the duty cycle of each cell is modulated in real time according to its SoC. During the balancing process, additional switching elements, such as L-bridge or H-bridge circuits, are strategically deployed at both the cell and pack levels to enable selective bypassing. In [18], a L-bridge configuration is employed within the bypass balancing system, as illustrated in Fig. 2. This approach enables cell balancing in large-scale battery packs by bypassing specific cells according to their SoC. The use of a half-bridge topology reduces the number of required components, thereby decreasing power losses and cost while enhancing efficiency and compactness.

The lack of bulky DC-DC converter elements (inductors and capacitors) and significantly shorter balancing time make this topology particularly well-suited for cell balancing in EVs. However, bypass strategies require additional circuitry for each cell, leading to increased cost and system complexity. Furthermore, as more cells are bypassed, the overall pack voltage decreases, which necessitates careful management.

2.1.2. CTC

Cell-to-cell balancing is a charge balancing technique that transfers excess energy from cells with higher charge levels to adjacent cells with lower charge levels. Energy redistribution in this method is achieved through the use of storage elements including inductors, capacitors, and converters. This technique is further categorized into ACTC and DCTC balancing.

ACTC

ACTC balancing methods achieve charge balancing by transferring energy between two adjacent cells, as illustrated in Fig. 3(a) [19–22]. The balancing speed of this technique is a function of both the cell count in the battery string and the spatial configuration of the imbalanced

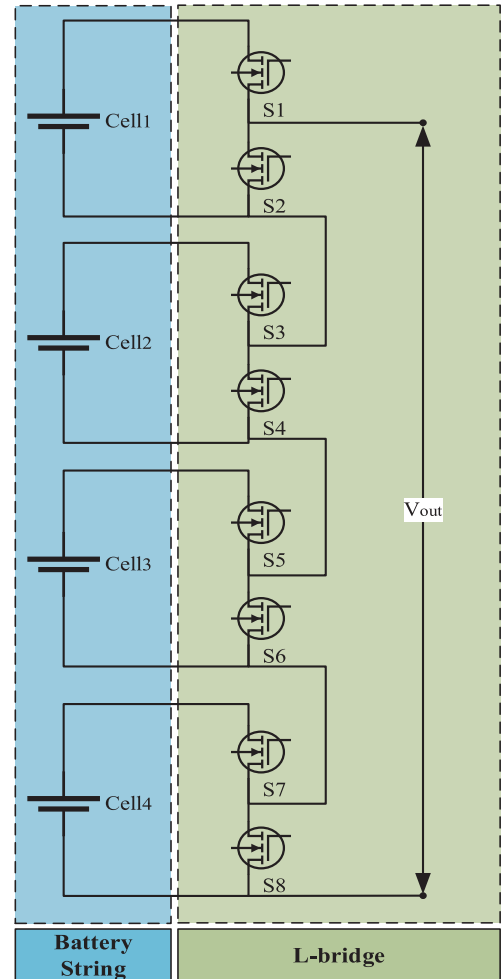


Fig. 2. L-bridge cell bypass CBC.

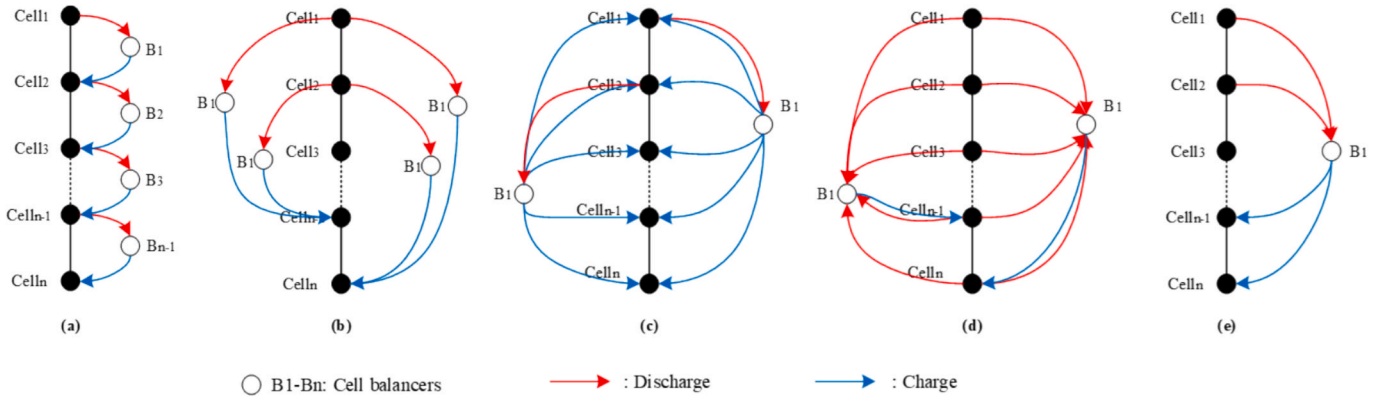


Fig. 3. Energy redistribution paths corresponding to various balancing methods, assuming Cell1, Cell2 are higher charged, Bn – 1 and Bn are lower charged and the other cells are in a balanced state. (a) ACTC (b) DCTC (c) CTP (d) PTC (e) MCTMC.

cells. Although the ACTC topology offers notable advantages such as control simplicity, modular scalability, and high system reliability, its performance can be significantly compromised when the source and target cells are positioned at opposite ends of the battery string. In such scenarios, energy transfer requires multiple intermediate stages, leading to prolonged balancing time and increased conversion losses. Consequently, the overall balancing efficiency and system performance are adversely affected under large spatial separations between the highly charged and the less charged cells.

DCTC

DCTC balancing, as proposed in [23–25], facilitates energy exchange between any two cells in a battery string, thereby eliminating the adjacency constraint of ACTC-based methods. The balancing pathways employed by the DCTC technique are depicted in Fig. 3(b). This method is implemented using a centralized converter topology augmented by a dynamic switching array for cell selection and control to enable energy transfer. DCTC offers higher balancing speed than ACTC but suffers from efficiency losses due to the large number of switches involved [26]. However, by employing a single balancer shared by all cells, the design remains compact and cost-efficient, making it well-suited for high-voltage battery systems.

CTC balancing achieves high efficiency and fast response in small-scale battery systems, benefiting from short energy transfer paths and low transmission losses. However, its practical implementation involves considerable control complexity, primarily because a large-scale switching network is essential to support configurable energy transfer paths among cells. Additionally, the single-path configuration limits energy exchange to only two cells per cycle, rendering the approach less suitable for high-capacity battery systems requiring scalable and parallel balancing mechanisms.

2.1.3. CTP

CTP balancing techniques operate by extracting excess energy from the most charged cell within the pack and distributing it uniformly across all cells through the pack terminals, as shown in Fig. 3(c) [27,28]. This iterative process continues until charge equilibrium is achieved, necessitating the integration of sensors and a controller to identify overcharged cells and manage the balancing operation. Benefiting from the high voltage difference between cells and the pack-level terminal, the CTP topology enables comparatively fast balancing. Its typical implementation involves power converters, multiple transformers, and multi-winding transformer designs. By using a shared pack bus, the CTP approach simplifies circuitry and reduces hardware costs. Nevertheless, energy must pass through multiple conversion stages within the pack voltage domain, leading to significant efficiency losses [29]. Additionally, the fixed voltage ratio between the pack and the cells imposes inherent limitations on balancing speed. Therefore, the CTP topology

demonstrates strong applicability in cost-sensitive, large-scale battery systems where compactness and modularity are prioritized over high efficiency.

2.1.4. PTC

In the PTC balancing strategy, energy is extracted from the battery pack and delivered to the least charged cell, ensuring charge balancing across the system. Fig. 3(d) demonstrates the specific energy transfer paths adopted by the PTC balancing scheme. While the PTC method shares many advantages with the CTP technique, the presence of multiple transformers or multi-winding transformer configurations within the balancing circuit introduces additional energy losses, which in turn limit the achievable efficiency. PTC-based balancing systems inherently support centralized control schemes, allowing coordinated SoC redistribution across the entire battery pack. This unidirectional energy flow simplifies control and is especially effective in restoring charge to deeply discharged cells, making the method well-suited for pack-level management in large-scale battery systems.

2.1.5. CTPTC

The CTPTC balancing method integrates the principles of both CTP and PTC to enhance power flow efficiency. During the charging, surplus energy originating from the most charged cell is actively redirected to the shared bus of the battery pack [30]. Conversely, during discharging, when a cell's SoC or voltage level falls below a specified reference threshold, the PTC mechanism enables energy transfer from the battery pack to the designated cell [29]. This method achieves a significantly higher balancing speed compared to standalone CTP and PTC techniques [6]. Nevertheless, its practical implementation is challenged by increased system size, elevated costs, and complex control requirements. The CTPTC configuration features enhanced architectural flexibility, as it enables the BMS to dynamically determine optimal energy balancing paths across the pack. This dynamic control not only prolongs the lifespan of the battery system but also ensures superior adaptability under complex and mixed-operating conditions.

2.1.6. MCTMC

The MCTMC topology enables direct energy transfer from consecutively more-charged cells to less-charged cells, as illustrated in Fig. 3(e), without the need for intermediate queuing [31]. This parallel energy transfer significantly accelerates the balancing process and suppresses redundant conversion losses. The operational efficiency of such a topology is linked to the design of the employed converter, which must support bidirectional energy flow and dual-mode conversion (boost and buck) to adapt to varying cell conditions. With proper implementation of Zero Current Switching (ZCS), the system can achieve rapid and efficient group-level balancing [32]. Nevertheless, the architecture inherently requires a complex switching matrix and advanced control algorithms to

manage the flexible energy routing. Compared to traditional topologies, this results in a larger component count, increased controller complexity, and elevated system cost. Consequently, MC2MC architectures are primarily recommended for high-performance applications demanding fast, efficient, and flexible balancing strategies, and are generally unsuitable for low-cost or space-constrained battery systems.

Table 1 offers a structured summary of the methods reviewed above.

2.2. Classification based on the components of energy transfer

According to the components utilized in the energy transfer process, active CBCs are classified into four categories: capacitor-based, inductor-based, DC-DC converter-based, and transformer-based configurations.

2.2.1. Capacitor-based balancing circuit

Capacitor-based CBCs employ an external capacitor to intermittently store and transfer energy between individual cells and the battery pack. The switching process is executed automatically by detecting voltage variations among cells, obviating the need for active current control. These circuits can be broadly categorized into three configurations: switched capacitor, single switched capacitor, and double-tiered switched capacitor architectures. As depicted in Fig. 4, switched capacitor CBCs achieve charge redistribution by cyclically switching capacitors between adjacent cells. When a voltage imbalance exists, capacitors sequentially charge from a higher-energy cell and discharge into a lower-energy adjacent cell, thereby equalizing the charge distribution. Owing to the indirect interconnection among all cells, the balancing strategy ensures progressive convergence toward balance

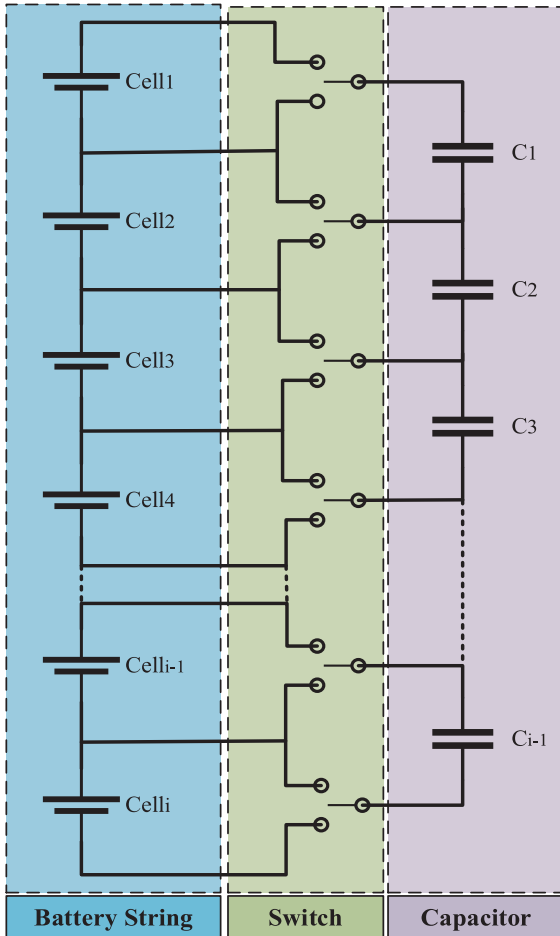


Fig. 4. Switched capacitor CBC.

throughout the battery pack. The major advantages of this approach include its inherent efficiency, relatively low circuit complexity, and applicability across a broad power range. Additionally, it does not require voltage sensing components or closed-loop control mechanisms. However, a significant drawback is the inherently slow balancing speed, as the magnitude of the balancing current is dictated by the voltage difference between adjacent cells. As the gap narrows, the balancing current reduces proportionally, thus increasing the duration required for balancing.

To address the trade-off between balancing efficiency and the constraints of physical size and cost, Shang [33] proposed a mesh-capacitor balancing topology capable of direct energy transfer between any two cells. This configuration supports modular design by linking the central nodes of individual mesh units, thus allowing deployment across multiple modules within a battery pack. Furthermore, the low voltage stress experienced by the MOSFETs enhances system reliability.

The double-tiered switched-capacitor equalizer is an advanced version of the traditional capacitor-based cell balancing circuits, employing two layers of capacitors to enable more efficient energy transfer between cells. This hierarchical structure achieves faster balancing by enabling parallel charge redistribution paths, effectively reducing the balancing time to one-fourth that of the traditional single-capacitor scheme. Despite this advantage, the topology introduces higher implementation costs and larger physical size, while the balancing speed still lags behind that of more advanced techniques. To improve performance further, Anurag [34] proposed a multi-layer balancing strategy that combines buck-boost conversion with the double-tiered switched-capacitor network. This configuration accelerates balancing and minimizes switching currents, offering a promising trade-off between complexity and performance.

In a single-capacitor switched-capacitor equalizer, energy transfer is conducted between a higher-SoC cell and a lower-SoC cell through a time-shared capacitor under a simplified control mechanism. This structure comprises one capacitor and a switch network, enabling DCTC balancing in a sequential manner. Its minimal component count allows for cost-effective scalability. Nonetheless, its primary limitation lies in the slow balancing speed due to the sequential operation and limited transfer bandwidth. To overcome this, La [35] proposed a fishbone-shaped switch-matrix topology, still based on a single capacitor, which leverages an optimal cell-pairing algorithm to determine switching actions dynamically. The addition of a current sensor further improves balancing performance by accelerating the balancing process and minimizing energy loss.

2.2.2. Inductor-based balancing circuit

Inductor-based CBCs constitute a category of charge balancing methods that rely on inductive components for energy conversion. These circuits enable the redistribution of energy from a higher-energy storage element—whether a cell, module, or battery pack—to a corresponding lower-energy unit within the system. The two principal configurations of this technique include single-inductor-based and coupled-inductor-based CBCs.

The single-inductor-based CBCs employs a single inductor to facilitate directed energy exchange between cells exhibiting energy surplus and those with energy deficit, as illustrated in Fig. 5. This method functions by continuously assessing individual cell voltage levels and dynamically selecting two cells to engage in energy exchange via a switched inductor mechanism. Due to its simplified structure and reduced number of components, this method offers advantages such as compact circuit design and lower system volume. However, a key drawback, analogous to that of the single-switched-capacitor balancing technique, is the substantial increase in balancing time when deployed in large-scale battery configurations with extensive series-connected cell strings. In Vardhan's [36] work, an automatic CBC was presented, featuring a single-switch inductor-based design governed by a SoC-based logic controller. To ensure stable energy transfer and enhance

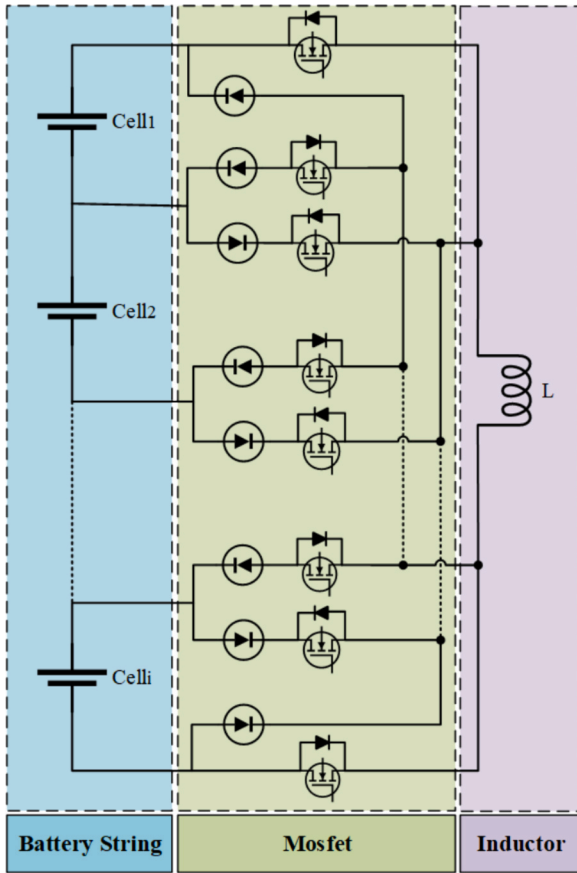


Fig. 5. Single-inductor-based CBC.

balancing speed, a constant inductor current is maintained using a PI controller.

Unlike the single-inductor balancing circuits, the coupled-inductor CBCs approach continuously analyzes the voltage state of adjacent cells within the battery pack and subsequently directs energy transfer from a highly charged cell to a lower-energy cell across the entire pack. This process is facilitated through precise control of MOSFET switches via pulse-width modulation (PWM) signals, enabling systematic energy redistribution. Although the coupled-inductor CBC achieves efficient charge balancing, its balancing time is inherently longer than that of the single-inductor CBC, especially in large battery packs, as energy must be sequentially transferred from the most charged cell to the least charged one. Despite its high balancing efficiency, this method is associated with magnetization losses, which can contribute to reduced overall system efficiency. In [37], Moghaddam introduced a coupled inductor-based CBC, in which all inductors share a single air-gapped magnetic core and are configured with optimized winding ratios. The proposed method offers advantages in terms of reduced hardware complexity and improved balancing speed, making it a promising alternative to conventional multi-inductor designs.

2.2.3. Transformer-based balancing circuit

Cell balancing circuits utilizing transformers are known for their high efficiency and fast balancing capabilities. Basic topologies operate in either CTP or PTC modes, while bidirectional switching combined with advanced control techniques enables MCTMC balancing. Despite their performance advantages, transformer-based circuits are associated with high costs, complex control requirements, and significant spatial demands. These circuits are generally classified into three categories: multiple transformer structures, switched single-transformer configurations, and multi-secondary winding transformer designs.

The multiple-winding transformer-based CBC assigns a dedicated transformer to each battery cell, with all primary windings connected in common and controlled by a single switch, as illustrated in Fig. 6. These transformers operate under PWM control to transfer excess energy from energy-surplus cells to energy-deficient ones, supporting high-speed balancing with reduced current and voltage stress on the switches. However, employing a dedicated transformer per cell significantly elevates implementation cost, system volume, and magnetization losses. Due to its structural characteristics, this design is compatible with large-format battery packs and modular configurations.

To minimize costs, a single transformer can be integrated with a switching topology similar to that of a single-switched inductor or capacitor. Through precise switch control, this configuration enables bidirectional energy transfer—either from the centralized bus to a low-SoC cell or from a high-SoC cell back to the bus—via the transformer. This balancing technique is widely employed in energy storage systems (ESS) to maintain voltage uniformity across cells and modules. However, the requirement for multiple switching components, along with the transformer, results in a relatively high implementation cost.

Combining the strengths of both preceding techniques, the CBC employing a multi-secondary winding transformer has been the subject of widespread investigation in the literature. This topology employs a single-core transformer featuring a primary winding and multiple secondary windings, each dedicated to a battery cell. Upon detecting an imbalance, various control strategies can be applied. Research efforts focus on optimizing transformer design to enhance performance while minimizing size and cost. Compared to the multiple-transformer approach, this configuration reduces the number of windings but often requires a custom multi-winding transformer, significantly increasing

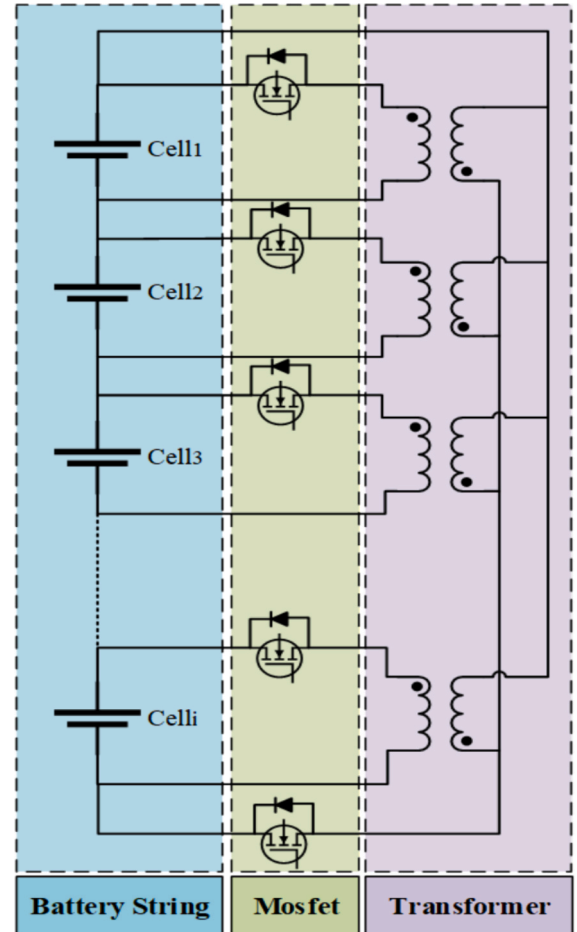


Fig. 6. Multiple-winding transformer-based CBC.

costs. A modular design approach can help mitigate these expenses.

In Shang's [38] work, a forward conversion-based global balancing framework was presented, enabling simultaneous realization of cell-level and module-level balancing, as well as demagnetization, in a single circuit. This integration significantly reduces system volume, cost, energy losses and implementation complexity. Moghaddam [39] enhanced the traditional multi-secondary transformer structure by introducing a full-bridge primary switching module, enabling zero-voltage switching (ZVS) operation without requiring an air gap. This modification results in reduced switching losses, shorter balancing time, and improved efficiency.

2.2.4. Converter-based balancing circuit

In contemporary electronic systems, DC-DC converters serve a critical function by regulating voltage levels in diverse applications, ranging from ESSs to EVs. These converters enable the bidirectional flow of energy between individual cells and the battery pack in a balancing system. The selection of an optimal DC-DC converter topology necessitates consideration of key parameters, including circuit complexity, galvanic isolation, power capacity, energy efficiency, and voltage gain. The component count has a direct impact on system complexity, implementation cost, and spatial footprint, whereas the conversion efficiency of the DC-DC converter critically affects the overall balancing performance, underscoring its importance in system-level design.

As key components in balancing systems, DC-DC converters are employed either to adjust voltage levels via step-up or step-down conversion or to mediate energy transfer between cells. The voltage conversion requirements differ based on the balancing topology: while PTC and CTP balancing necessitate converters with a high voltage conversion ratio, CTC configurations can achieve efficient operation with converters that have a relatively low voltage conversion ratio.

DC-DC converters are commonly classified based on their energy transfer directionality, which can be either unidirectional or bidirectional. Unidirectional converters permit energy flow solely from the source to the load, whereas bidirectional converters support energy exchange in both directions. Furthermore, DC-DC converters are classified as either isolated or non-isolated, depending on whether a transformer is employed in the circuit. Isolated converters incorporate high-frequency transformers to achieve galvanic isolation between the primary and secondary sides. Nevertheless, their intrinsic limitations, including high-voltage stress, larger system volume, increased number of switching elements, and higher overall cost, must be considered during system design.

The advantages and disadvantages of different component-based active cell balancing topologies are systematically presented in Table 2.

3. Classification based on DC-DC converter

This section explores the fundamental topologies of DC-DC converters. Non-isolated converters, including buck-boost, Ćuk, and SEPIC/Zeta, do not provide galvanic isolation between the source and the load, making them structurally simpler. In contrast, isolated converters utilize an isolation transformer to electrically separate the input from the output, with flyback, forward, dual active bridge (DAB), dual half-bridge (DHB), resonant, and push-pull topologies being prominent examples. This section elaborates on various DC-DC converter architectures employed in cell balancing strategies.

3.1. Non-isolated converters

3.1.1. Buck-boost converter

The buck-boost converter-based CBC comprises an inductor and two power MOSFETs with integrated body diodes, as depicted in Fig. 7. Its operation is based on energy transfer from a higher-energy cell to an adjacent lower-energy cell, achieved through switching control that responds to the detected voltage difference between neighboring cells.

Table 2

Advantages and disadvantages of different component-based active cell balancing topologies.

Components of energy transfer	Advantages	Disadvantages	Application
Capacitor-based	Simple structure; low cost; no voltage sensing or closed-loop control required	Low balancing speed, efficiency drops with small voltage differences	Small-scale systems with low accuracy and speed requirements
Inductor-based	Compact design; fewer components; adaptable to various SoC control schemes	Balancing time increases with more cells, suffer from magnetization loss, sequential energy transfer	Medium-scale packs requiring moderate performance and simple design
Transformer-based	Supports galvanic isolation; high energy transfer accuracy; suitable for bidirectional flow	Large physical size, high cost, complex control, not easily scalable	High-voltage or safety-critical applications needing isolation
Converter-based	High flexibility; supports various control strategies and energy routing	High complexity, may involve many switches and control circuits, possible cost and EMI concerns	Advanced battery systems needing flexible, high-speed, and intelligent balancing

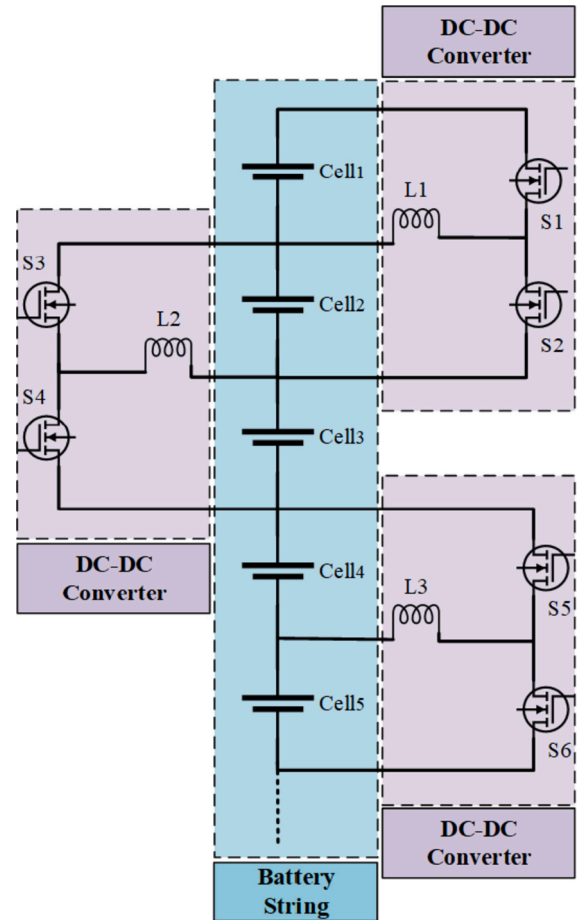


Fig. 7. Buck-boost converter-based CBC.

As one of the most prevalent topologies in ACTC balancing systems, the buck-boost converter is widely employed due to its ability to balance individual cells and its straightforward implementation. In a battery pack consisting of N cells, $N-1$ buck-boost converters are required. The voltage across each converter does not exceed the maximum cell voltage, resulting in low power dissipation. However, the need for numerous converters leads to slow balancing speed, increased cost, and larger system size.

To improve the performance of conventional buck-boost converters, researchers have explored various structural modifications. In [40], an isolated buck-boost topology incorporating a transformer is proposed to provide galvanic isolation between cells, effectively isolating faults and enhancing system protection. Xu [41] presents a non-isolated balancing method that achieves efficient redistribution of energy among cells without introducing supplementary circuit elements. Furthermore, a marginal duty cycle control scheme is adopted to accelerate the balancing process, achieving a faster balancing speed than the conventional ACTC equalizer. Manjunath [42] develops a modular two-stage balancing architecture employing an enhanced buck-boost converter. By combining cell-level and module-level balancing, the system significantly reduces the overall balancing time during dynamic load conditions while maintaining hardware simplicity.

3.1.2. Ćuk converter

The Ćuk converter achieves efficient CTC balancing by utilizing a capacitor as the intermediary energy transfer component, distinguishing it from the buck-boost converter, which depends on an inductor for energy transmission. In this approach, charge is temporarily stored in a capacitor sourced from high-SoC cells and then delivered to neighboring low-SoC counterparts. According to Fig. 8, the Ćuk converter is composed of an energy transfer capacitor C , two independent inductors L_1 and L_2 , and two MOSFETs M_{11} and M_{12} , both incorporating body diodes. The balancing process is governed by a straightforward 50 %

duty cycle applied to the switches, facilitating energy redistribution among cells until equilibrium is reached. The inclusion of inductors enables stable current regulation to the capacitor across both energy storage and release cycles, offering improved efficiency over switched-capacitor configurations. Additionally, this topology supports bidirectional energy conversion with variable voltage gain, irrespective of the input–output polarity. The Ćuk converter-based balancing topology offers all the benefits of ACTC charge transfer while using cost-effective capacitors for energy storage. It also ensures minimal ripple in input and output currents. Nevertheless, its large circuit volume and sequential balancing nature limit balancing speed, as energy must pass through each cell to balance the entire battery pack.

In Ling's [43] work, a bidirectional isolated Ćuk converter was introduced, featuring capacitive isolation between each cell and the energy bus, and leveraging synchronous switching control to enhance energy balancing performance. However, the design inherits complexity and loss issues typical of traditional Ćuk converters. A modified Ćuk-based equalizer was proposed by Moghaddam [44], wherein the number of switches scales proportionally with the cell count n , instead of the conventional $2n - 2$ configuration. This reduction in component count leads to lower conduction and switching losses, thus improving the overall system efficiency. He [45] further enhances the balancing performance by combining a soft-switching technique with a ZCS-enabled bidirectional Ćuk converter, leading to lower switching loss and a marked improvement in the efficiency of the balancing mechanism.

3.1.3. SEPIC/Zeta converter

The SEPIC (Single-Ended Primary-Inductor Converter) and Zeta are both non-isolated DC–DC converters derived from a reconfiguration of the components in the Ćuk converter, enabling a positive, non-inverting output voltage. Their structural difference lies in the placement of the

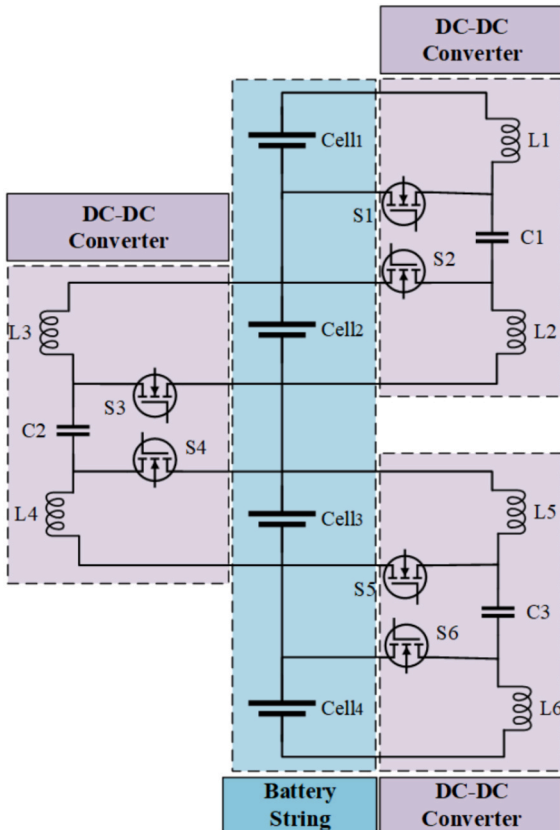


Fig. 8. Ćuk converter-based CBC.

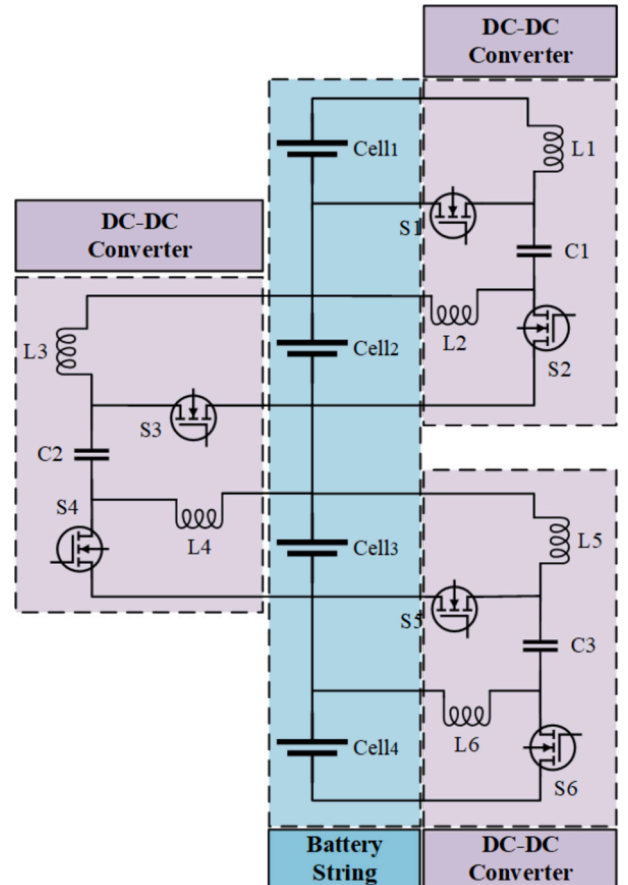


Fig. 9. SEPIC converter-based CBC.

input and output inductors around the coupling capacitor, while both achieve high efficiency across a wide input voltage range. Similar to the Ćuk converter, the SEPIC employs an capacitor-based energy transfer mechanism, making it suitable for CTC balancing, as depicted in Fig. 9. Wu [46] presented a cell balancing strategy incorporating fuzzy logic for grouping-based balancing, allowing energy redistribution between individual cells and cell groups. Compared to traditional methods, it demonstrates reduced balancing time and energy loss, particularly during dynamic charge/discharge cycles. Notably, this work emphasizes advancements in control strategies rather than changes in circuit topology.

Although the SEPIC topology alone is not a mainstream choice for battery balancing, recent studies have combined SEPIC and Zeta converters to realize bidirectional two-level equalizers with step-up/step-down capability and reduced switch count. Liu [47] proposed a layered balancing strategy considering both single batteries and battery cells, thereby improving efficiency and accelerating balancing. Liao [48] further employed bidirectional SEPIC–Zeta converters to achieve multi-scale energy redistribution, enabling energy transfer both within and across groups while enhancing balancing performance.

3.2. Isolated converters

3.2.1. Flyback converter

As a topological derivative of the buck-boost converter, the flyback topology provides galvanic isolation in DC-DC conversion. Instead of a discrete inductor for energy storage, it employs the magnetizing inductance inherent to the transformer to fulfill this function. A flyback converter-based CBC is commonly designed with an integrated switch array to identify and select either undercharged or overcharged cells. The relatively low number of circuit components in this topology simplifies the overall circuit design. Furthermore, by leveraging bidirectional switching and control, the flyback structure enables multiple energy transfer modes, including CTP, PTC, and CTPTC. Flyback topology supports dual-mode operation, encompassing both unidirectional and bidirectional energy transfer. In a unidirectional design, a single switch governs energy transfer, whereas in a bidirectional implementation, two switches are employed to allow energy flow in both directions.

Zabetian [49] introduced a unidirectional flyback converter as a CBC operating in CTP mode, which provides a speed advantage through the concurrent activation of all cells within the balancing scheme. By adopting a multi-winding input–output flyback topology, the system significantly improves energy transfer efficiency. However, compared to the single-winding converter, it demands more space, potentially increasing the overall system size and weight. Additionally, the presence of multiple windings can introduce higher leakage inductance and elevated costs.

A novel PTC-mode topology with a simple structure is proposed in [50]. This design incorporates a unidirectional flyback converter to enable direct energy exchange between the entire battery system and a targeted cell, effectively minimizing the transfer distance and accelerating the balancing process. Furthermore, the system exhibits high scalability, as the number of MOSFETs can be adjusted accordingly to accommodate variations in pack cell configuration, facilitating ease of expansion and implementation.

Based on the work presented in [50], Guo [51] proposed a structurally optimized and reliable CTPTC-mode topology. A pair of unidirectional flyback converters is incorporated in this topology to independently perform PTC and CTP operations, with the switching matrix enabling appropriate routing of energy. While the design effectively simplifies control complexity, the requirement of two transformers leads to an increase in overall system size. To further improve integration and compactness, Cui [52] implemented a bidirectional flyback converter to facilitate coordinated CTPTC balancing, as depicted in Fig. 10. This topology enhances energy transfer reliability by

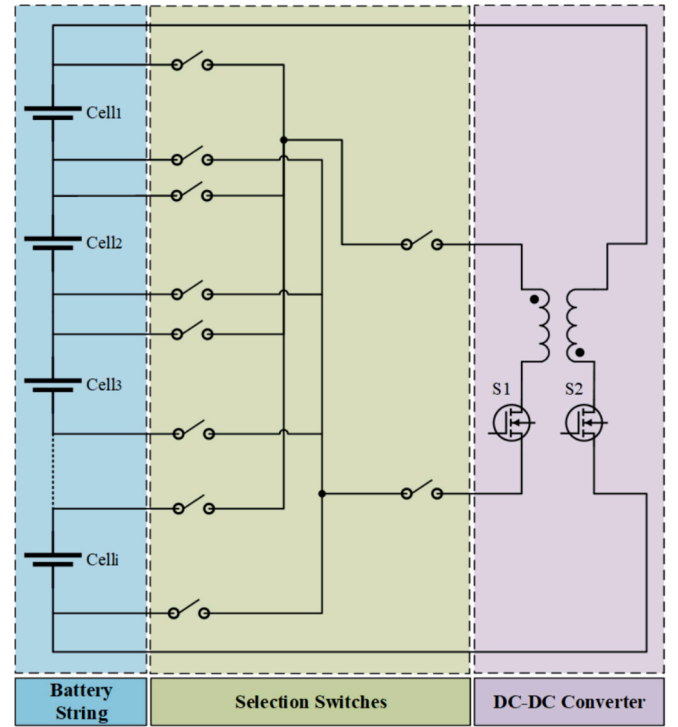


Fig. 10. Flyback converter-based CBC.

accommodating both charge and discharge modes. Although the adoption of a bidirectional flyback-based balancing topology simplifies the topology structure, the control strategy is more complex.

3.2.2. Forward converter

As a representative magnetically coupled DC-DC converter, the forward converter supports bidirectional voltage conversion, depending on the transformer turns ratio. While its output characteristics resemble those of a buck converter, the presence of a transformer introduces voltage scaling and galvanic isolation, which are absent in non-isolated topologies. The operation of the forward converter is governed by a high-frequency ON-OFF switching cycle, controlled by PWM or other regulation circuit. During the ON interval, the primary winding is energized, establishing a magnetic field in the core. When the switch transitions to the OFF state, the magnetizing current ceases, causing the collapsing magnetic field to induce a voltage across the secondary winding, which delivers energy to the output. Distinct from the flyback converter, the forward converter allows for immediate power delivery during the switch conduction phase. This operating principle facilitates continuous output current, reduced output voltage ripple, and superior dynamic response. In battery management systems, the forward converter has been successfully applied to implement unidirectional balancing strategies such as PTC and CTP. Nevertheless, the realization of bidirectional energy transfer required for CTPTC balancing imposes additional design constraints. Specifically, bidirectional operation demands a more complex topology and advanced control algorithms, thereby increasing system-level complexity.

The balancing strategy proposed by Liu [28] utilizes a simplified CTPTC configuration with a bidirectional forward converter, as illustrated in Fig. 11, enhancing flexibility in cell balancing while maintaining structural simplicity. Zong [53] further advanced balancing circuit performance by integrating a resonant reset in the forward converter, enabling zero-voltage switching and supporting MCTMC balancing with higher speed and lower switching losses.

Although the forward converter provides robust controllability, low output ripple, and reduced electromagnetic interference, these improvements entail a trade-off in the form of elevated hardware

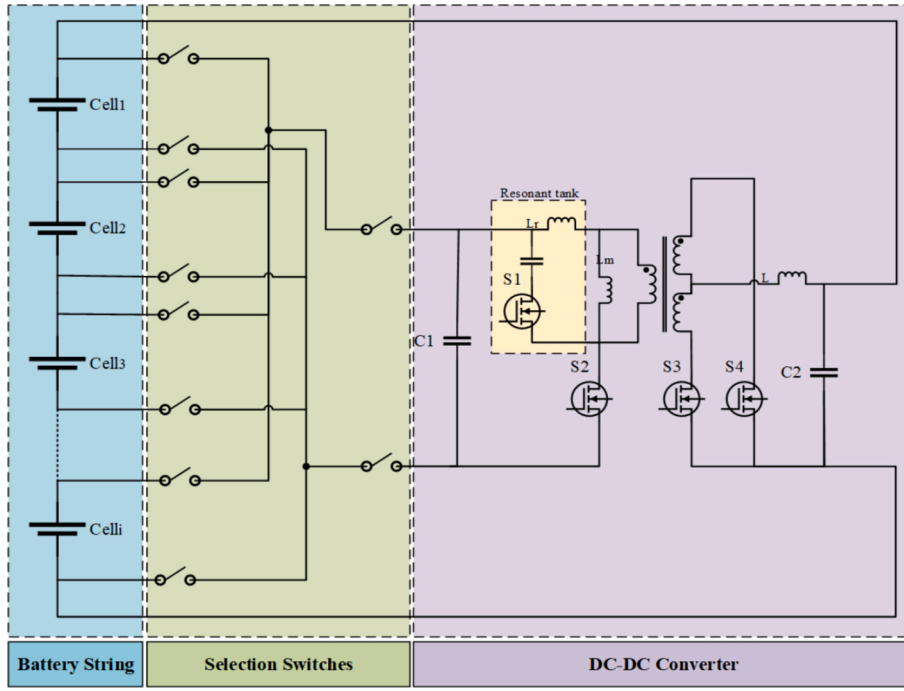


Fig. 11. Forward converter-based CBC.

complexity. Compared to the flyback converter, it necessitates additional circuit components for magnetic flux resetting and current routing, which results in higher conduction losses. Consequently, the overall balancing efficiency may be inferior under certain conditions, particularly when operating in low-power regimes.

To address the limitations of conventional isolated balancing topologies, recent research has explored hybrid configurations. Moghaddam's [54] approach integrates a centralized buck converter with multiple forward converters, minimizing hardware complexity while supporting multi-pack energy redistribution. Gao [55] developed a hybrid equalizer incorporating both bidirectional flyback and forward converters, enhancing balancing flexibility and performance.

3.2.3. Dual active bridge converter

Widely employed in bidirectional isolated energy conversion applications, the DAB converter comprises a pair of H-bridge circuits connected through a high-frequency transformer with intentionally designed leakage inductance, as shown in Fig. 12. Each H-bridge comprises four actively controlled semiconductor switches, enabling independent voltage regulation and full bidirectional operation.

The DAB converter offers several notable advantages, including galvanic isolation, high energy transfer capability, wide voltage conversion range, inherent modularity, and the potential for soft-switching. The achievement of ZVS in this topology results from the combined effects of phase-shifted control between the H-bridges and the resonant behavior introduced by the leakage inductance and inherent parasitics of the circuit. This results in reduced switching losses under

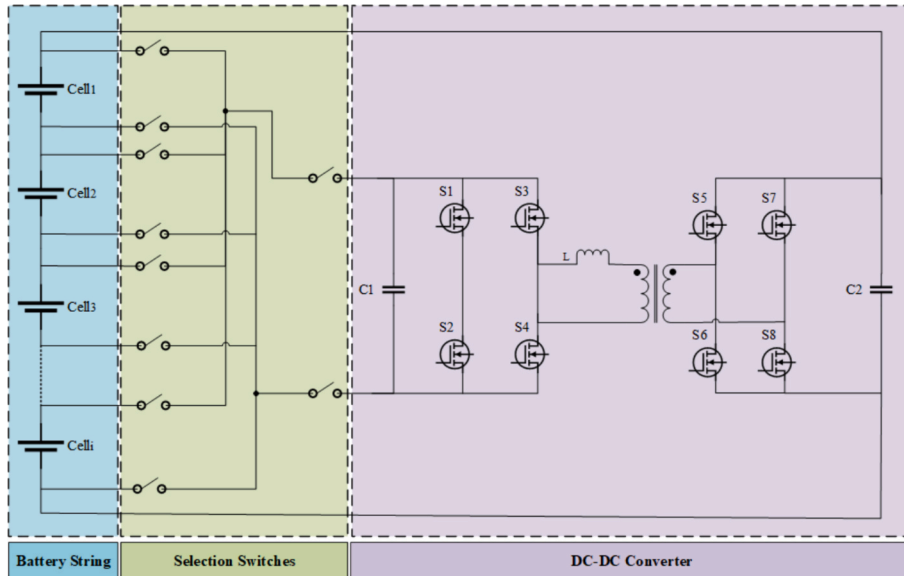


Fig. 12. Dual active bridge converter-based CBC.

appropriately selected operating conditions. Furthermore, this topology supports bidirectional operation, functioning in both step-up and step-down modes, which facilitates its application in a variety of battery management scenarios such as PTC, CTP and CTPTC balancing architectures. The scalability of power transfer with respect to the number of switches, under the condition of identical switch ratings, makes the DAB topology particularly favorable for deployment in high-power scenarios. However, the DAB converter also presents challenges. Under light load conditions, the failure to maintain ZVS leads to increased switching losses. Under heavy load conditions, the presence of high circulating currents results in elevated conduction losses. These factors impose practical limitations on the converter's efficiency and must be addressed in system-level design and control strategy development.

Bae [56] presents an SoC-based active balancing strategy utilizing DAB for real-time power flow control among battery cells. The approach improves battery utilization by mitigating effects of cell aging and reducing energy transfer losses. Ramesh [57] developed a single-input multi-output CBC with interleaved phase-shift control and a loss-minimizing algorithm. Given the popularity of DAB converters in isolated balancing systems, Ref. [58] provides a detailed review of DAB-derived topologies. However, the increasing diversity of applications and performance demands has exposed the limitations of traditional DAB schemes, prompting numerous structural enhancements and modulation innovations.

3.2.4. Dual half-bridge converter

The DHB converter, formed by integrating two half-bridge stages through a transformer, offers a simplified alternative to the DAB by halving the number of active switching components, as shown in Fig. 13. The incorporation of a capacitive voltage divider at the input reduces the output voltage to approximately half that of a DAB. The DHB topology is characterized by low ripple current, inherent ZVS capability, and efficient continuous power transmission with negligible circulating current, which makes it applicable to high-voltage balancing systems requiring bidirectional power flow. Despite these merits, the DHB converter is prone to elevated circulating current and switching losses under light load operation, which constrains its practical application.

In pursuit of high-frequency and compact balancing, Chen [59]

proposed a DHB equalizer utilizing a phase-shift control strategy, which enables efficient bidirectional energy transfer within and across cell groups. Wang [60] proposed a cost-efficient DHB topology for EVs, simplifying circuit design by minimizing switch count and adopting a coreless transformer. Zilio [61] introduced a multi-active half-bridge converter integrated with a multi-winding transformer to realize versatile cell balancing. The proposed topology supports multiple operation modes and allows energy redistribution between any two cells or between a cell and the entire pack, thereby enhancing balancing flexibility and circuit simplicity.

3.2.5. Resonant converter

Resonant converter-based balancing circuits utilize a resonant tank to both drive switching devices and transfer energy between battery cells of unequal charge levels. This approach circumvents the need for complex digital control systems. The core component, a resonant tank comprising a series or parallel combination of inductance and capacitance, ensures soft-switching operation by limiting transient voltage and current during switching events. A parallel resonant converter is illustrated in Fig. 14. These converters are categorized by order, including second-order (two-element), third-order (three-element), and higher-order variants. The second-order tank, typically consisting of a single inductor and capacitor, is foundational. Among the various topologies, LLC and CLLC resonant converters have shown the most promise for electric vehicle power systems, offering high efficiency, soft-switching capability over a broad load range, electrical isolation, and reduced electromagnetic interference. However, the benefits of resonant converters are accompanied by significant trade-offs, including more complex control requirements and increased cost associated with system design and implementation. Despite these challenges, resonant topologies remain widely studied and adopted due to their advantageous characteristics.

Pham [62] leverages an LLC resonant converter in a CTC configuration, ensuring high-speed balancing and soft-switching characteristics through a relatively simple control structure. In contrast, Shang [32] introduces an LC tank-based architecture enabling MCTMC balancing, which, despite its high speed and scalability, incurs substantial cost and size owing to the substantial count of required switches or relays.

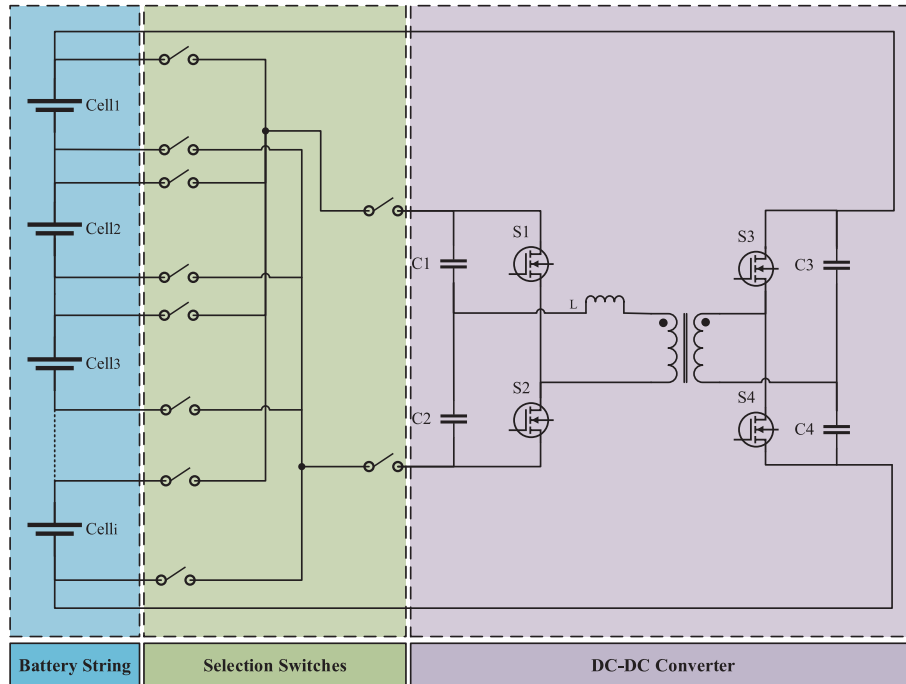


Fig. 13. Dual half-bridge converter-based CBC.

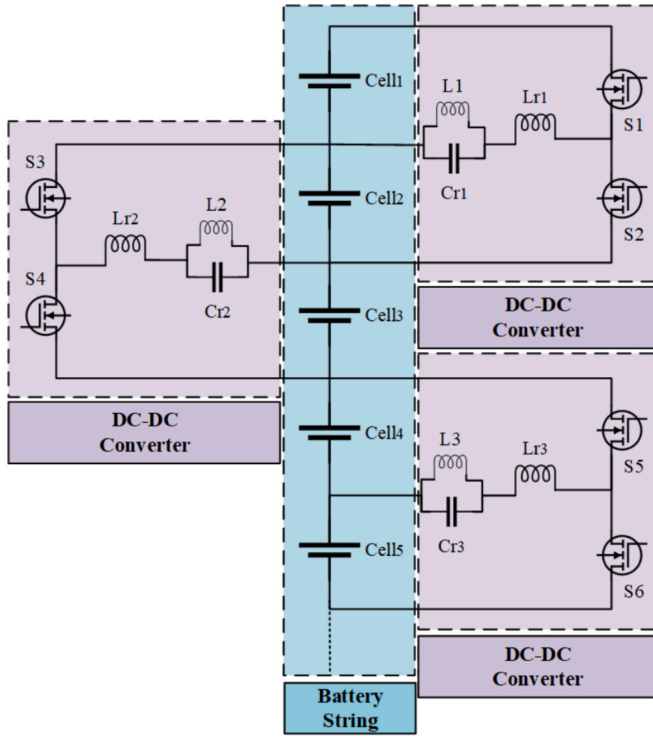


Fig. 14. Resonant converter-based CBC.

Expanding the scope to PTC balancing, Wei [63] employs an LCC multi-resonant design with open-loop constant-current control, allowing shared hardware for energy redistribution from the string to a target cell. Subsequently, Wei [64] introduced an improved bidirectional L2C3 resonant converter, integrating constant-current control and fixed-frequency operation. The proposed architecture enables CTPTC balancing, while maintaining efficient and consistent balancing performance.

3.2.6. Push-pull converter

The push-pull converter, structurally based on the buck converter, as illustrated in Fig. 15, performs both voltage conversion and electrical isolation using a center-tapped transformer. Each half of the winding conducts alternately, allowing the core to be magnetized in both directions and improving transformer utilization. Voltage control is based on the transformer turns ratio, following a feed-forward scheme. Push-pull converters are available in unidirectional and bidirectional forms. The bidirectional version, composed of four switches and a transformer, shares similar working principles with DAB and DHB converters. This converter is well-adapted to applications that demand galvanic isolation and inter-level voltage conversion.

In Pham [65], a high-speed CTC energy balancing approach was introduced, employing a single push-pull converter to facilitate direct energy exchange between voltage-imbalanced cells. Furthermore, push-pull converters can also be combined with other topologies for cell balancing. For example, Ref [66] proposed a soft-switched bidirectional DC-DC converter integrating a phase-shifted full bridge and a push-pull stage, achieving high efficiency and effective balancing in EV Li-ion battery strings.

3.3. Comparison of relevant studies

A comparison of representative DC-DC converter-based CBCs from existing literature is summarized in Table 3, evaluating key performance metrics including balancing speed, implementation cost, efficiency, and control complexity. The reviewed studies employed varying cell counts and initial difference, both of which significantly influence balancing dynamics. To enable meaningful cross-study comparison, the maximum initial voltage or SoC variance is specified for each case.

The analysis reveals characteristic trade-offs among different balancing approaches. The ACTC/DCTC topologies offer modular implementation and simple control through buck-boost, Ćuk, and SEPIC-based circuits, with competent balancing performance, yet suffer from linear hardware scaling. The CTP/PTC configurations demonstrate faster balancing through flyback, forward, and bridge converter implementations, suitable for large-scale systems, but incur significant space and cost penalties from magnetic components. Resonant converter

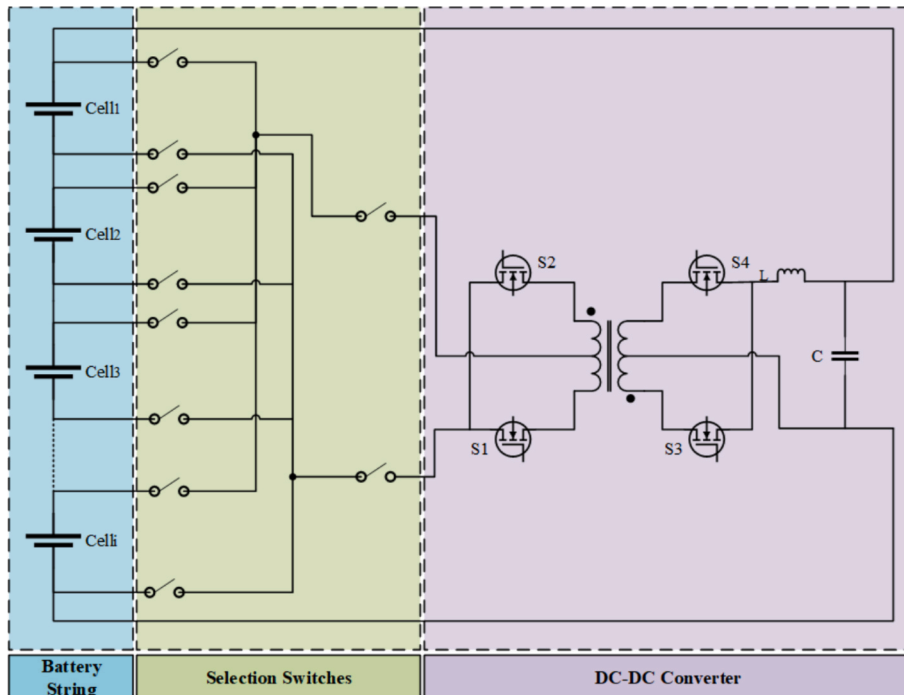


Fig. 15. Push-pull converter-based CBC.

Table 3

Comparison of active CBCs employing DC-DC converters.

Ref.	DC-DC converter	Control variable	Cell number	Max. Initial difference	Current path	Speed	Cost	Efficiency	Control complexity
[67]	Buck-boost	SoC	8	20.5 %	ACTC	1150 s	High	97.37 %	High
[68]	Ćuk	SoC	4	11 %	ACTC	2286 s	High	87.8 %	High
[48]	SEPIC	SoC	8	7 %	DCTC	1143 s	Low	Not given	Low
[69]	Flyback	SoC	7	15 %	PTC	2320 s	Medium	Not given	Medium
[53]	Forward	Cell voltage	8	0.042 V	MCTMC	1260 s	Medium	83.31 %	High
[70]	DAB	SoC	8	10 %	DCTC	~1500 s	High	Not given	High
[59]	DHB	Cell voltage	4	0.07 V	MCTMC	720 s	High	Not given	High
[64]	Resonant	Cell voltage	4	0.186	CTPTC	3610 s	Medium	90.1 %	High
[65]	Push-pull	Cell voltage	12	0.18 V	DCTC	3240 s	Medium	89.5 %	Medium

topologies, employing ZVS and ZCS soft-switching techniques, effectively mitigate switching losses but introduce considerable design complexity.

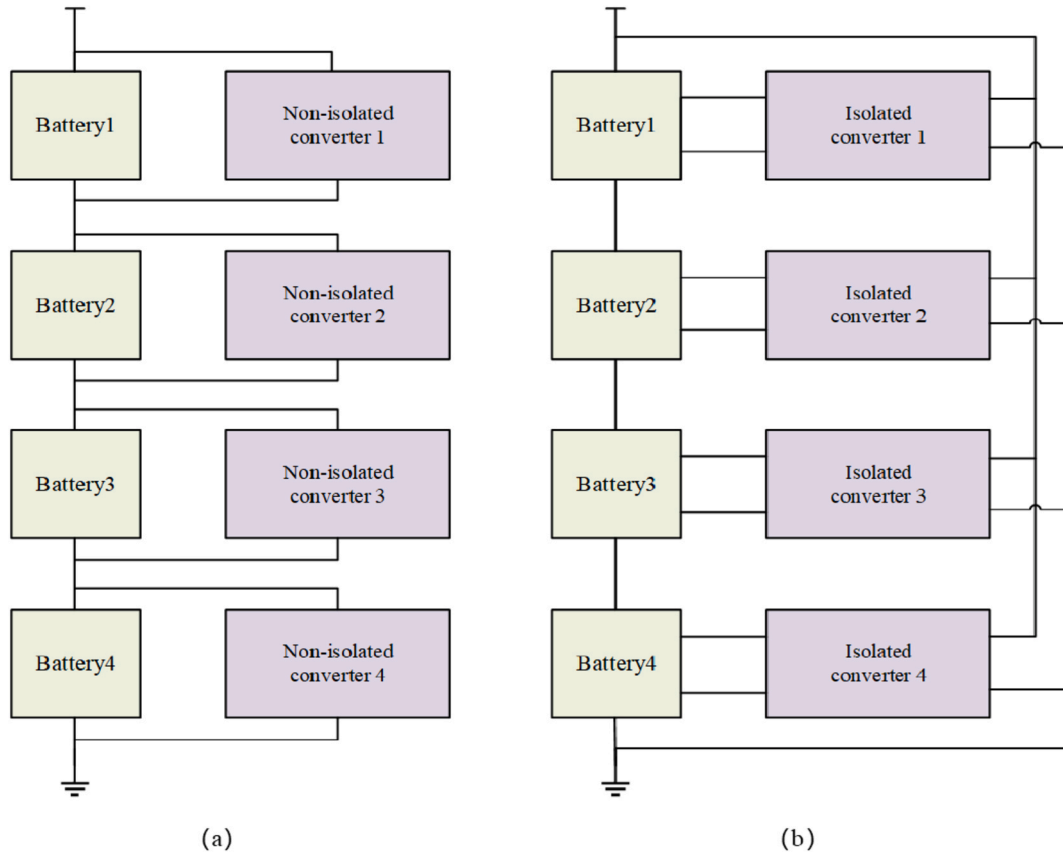
4. Comparison and discussion of four DC-DC converters

A comparative assessment of four DC-DC converter-based CBCs is conducted in this section, focusing on balancing speed and current ripple characteristics. The selected topologies include three non-isolated bidirectional converters (buck-boost, Ćuk, and SEPIC) and one isolated bidirectional converter (flyback). The non-isolated converters employ the circuit topology depicted in Fig. 16(a), which are capable of realizing CTC energy transfer. In contrast, the isolated converter utilizes the configuration shown in Fig. 16(b), which enables CTPTC energy redistribution.

The performance of cell balancing systems is determined by several interdependent factors, including balancing time, energy efficiency, control complexity, and power dissipation. Among these, the balancing speed is often regarded as a principal metric due to its inherent connection with both system efficiency and energy loss. While numerous

studies have thoroughly evaluated these conventional performance indicators of cell balancing architectures, current ripple remains an underexplored parameter, with no dedicated comparative analysis reported in the literature. Yet, current ripple plays a critical role in both the operational effectiveness and battery longevity. Specifically, the interaction between ripple current and the intrinsic impedance of battery cells results in increased conduction losses, thereby undermining the overall energy efficiency. Furthermore, sustained exposure to high-frequency ripple currents can lead to electrochemical polarization and thermal stress, accelerating capacity degradation and the progressive rise of internal resistance [71]. Fig. 17 illustrates the underlying mechanisms through which ripple current impacts battery efficiency and lifespan. Given its significant impact, a comparison of current ripple characteristics across different balancing topologies is warranted to better inform the design of high-performance battery management systems.

The active CBCs are modeled in MATLAB/Simulink for a battery pack comprising four series-connected lithium-ion cells. The nominal voltage and capacity of each cell are 3.7 V and 3.5 Ah, respectively, while the initial SoC values differ across the cells, as summarized in

**Fig. 16.** Converter-based CBCs.

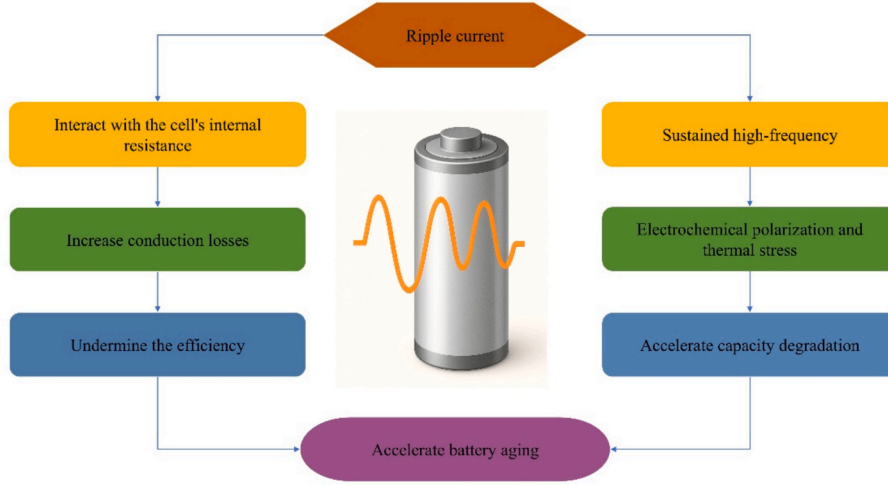


Fig. 17. Impact of ripple current on battery.

Table 4

Simulation parameters for four different DC-DC converter topologies.

Simulation parameters	Values
Cell ₁ (SoC%)	82
Cell ₂ (SoC%)	73
Cell ₃ (SoC%)	68
Cell ₄ (SoC%)	62
Nominal voltage	3.7 V
Nominal capacity	3.5Ah
Inductor (L)	100μH
Filter Capacitor (C)	50μF
Coupling Capacitor (C)	10μF
Charging and discharging current	2A

Table 4. For consistency, all converters operate with a fixed balancing current of 2A, and identical converter parameters and control algorithms are used across the configurations to ensure an unbiased performance comparison.

The effectiveness of active cell balancing strongly depends on the accuracy of SoC estimation. Inaccurate SoC estimation may lead to misjudgments in energy redistribution, thereby lowering efficiency and accelerating cell degradation [72]. In this study, SoC estimation is performed using the Coulomb counting (CC) method, as shown in Eq. (1), calibrated through the OCV–SoC fitting relationship.

$$SoC(t) = SoC_0 + \frac{1}{Q_{max}} \int_0^t i(t)dt \quad (1)$$

Where,

SoC_0 = initial SoC,

Q_{max} = maximum capacity,

$i(t)$ = current going in and out of a cell

However, the current approach assumes ideal conditions without incorporating dynamic error correction. For practical applications, additional filtering and compensation techniques—such as Extended Kalman Filters (EKF) or drift-corrected Coulomb counting—are required to improve robustness. Furthermore, precise voltage measurement with minimal noise is essential to prevent erroneous balancing decisions. Future developments are expected to integrate these techniques into hardware implementations to enhance system reliability.

4.1. Balancing speed comparison

To evaluate system behavior under varying operating conditions, both charging and discharging scenarios are simulated for each balancing topology. Fig. 18(a)–(h) provides a detailed illustration of the balancing behaviours of the battery pack under the four evaluated topologies. The results indicate that the flyback converter achieves the fastest balancing, completing the process in 62 s/67 s during charging/discharging. The buck-boost topology follows closely, with balancing durations of 64 s/73 s for charging/discharging. In contrast, the Ćuk converter exhibits the slowest performance, requiring 201 s for both charging and discharging cases.

To further compare the performance of the buck-boost and flyback topologies, the series configuration of the battery pack was expanded to include eight cells and their balancing behaviours are shown in Fig. 19. The balancing times for all studied configurations are summarized in Table 5. The findings confirm that the CTPTC scheme utilizing a bidirectional flyback converter maintains higher balancing speed than the buck-boost CTC approach under increased pack size.

4.2. Current ripple comparison

The structural differences among various DC-DC converter topologies lead to distinct numbers and arrangements of reactive components, resulting in significant variation in current ripple. Because the converter's output current directly interacts with the battery cells in the balancing circuit, this research is centered on comparing the peak output current ripple of each topology. The current ripple characteristics corresponding to the four converter topologies are illustrated in Fig. 20. Simulation results indicate that the Ćuk converter exhibits the lowest ripple due to its use of inductors on both the source and load sides, which ensures a continuous current path and allows the output inductor to effectively filter high-frequency switching components. The buck-boost and SEPIC converters demonstrate intermediate ripple levels, whereas the flyback converter shows the highest output current ripple.

4.3. Discussions about balancing speed and current ripple of four DC-DC converters

The Table 6 consolidates the aforementioned simulation results, from which the following key observations can be drawn from the comparison of balancing speed and current ripple.

In the context of bidirectional DC-DC converter-based CBCs, the non-isolated CTC topology exhibits varying balancing speeds depending on the converter type. Among non-isolated CTC configurations, the buck-

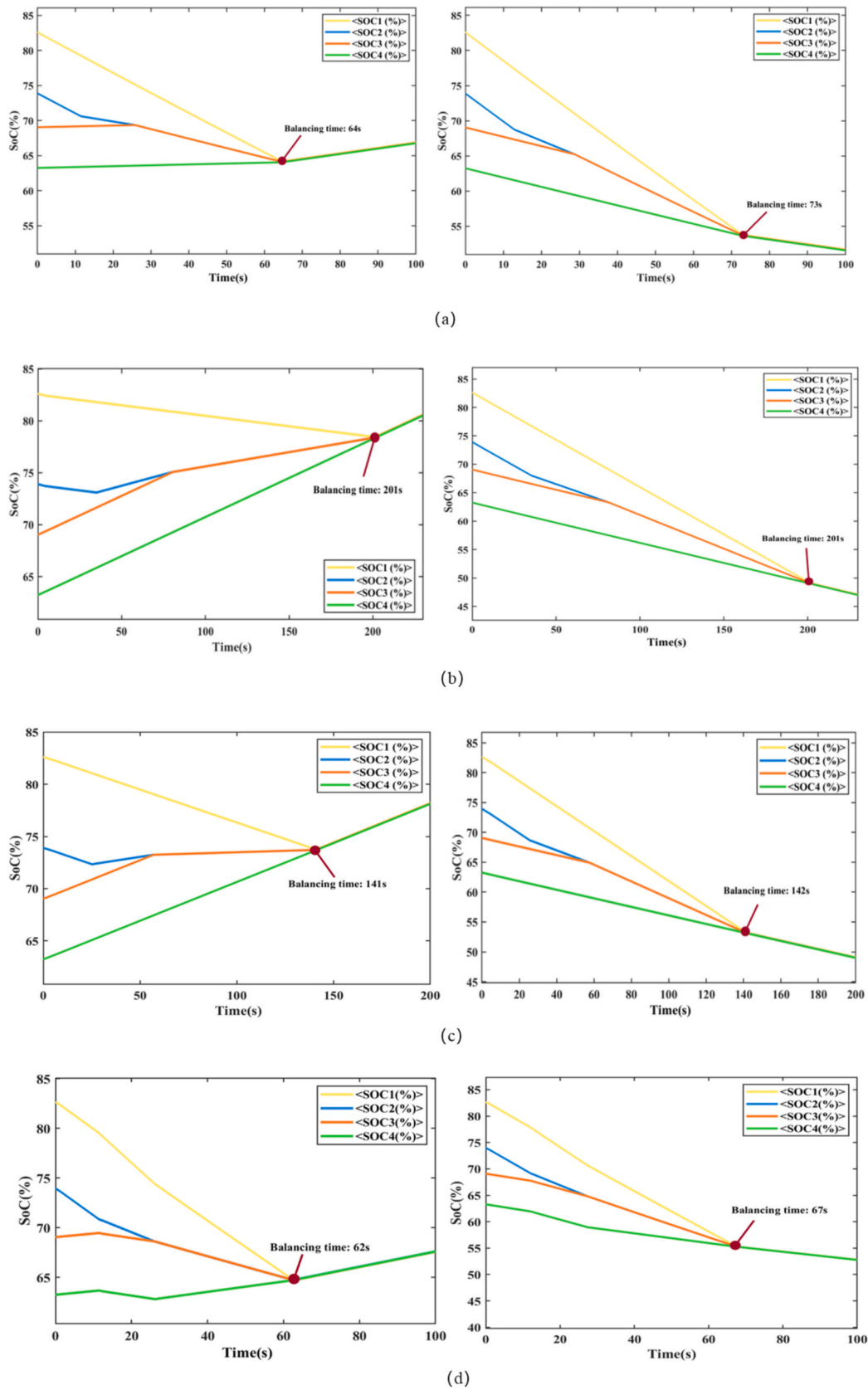


Fig. 18. Simulation results of balancing speed for four cells under charging and discharging conditions. (a) Buck-boost, (b) Ćuk, (c) SEPIC, (d) Flyback.

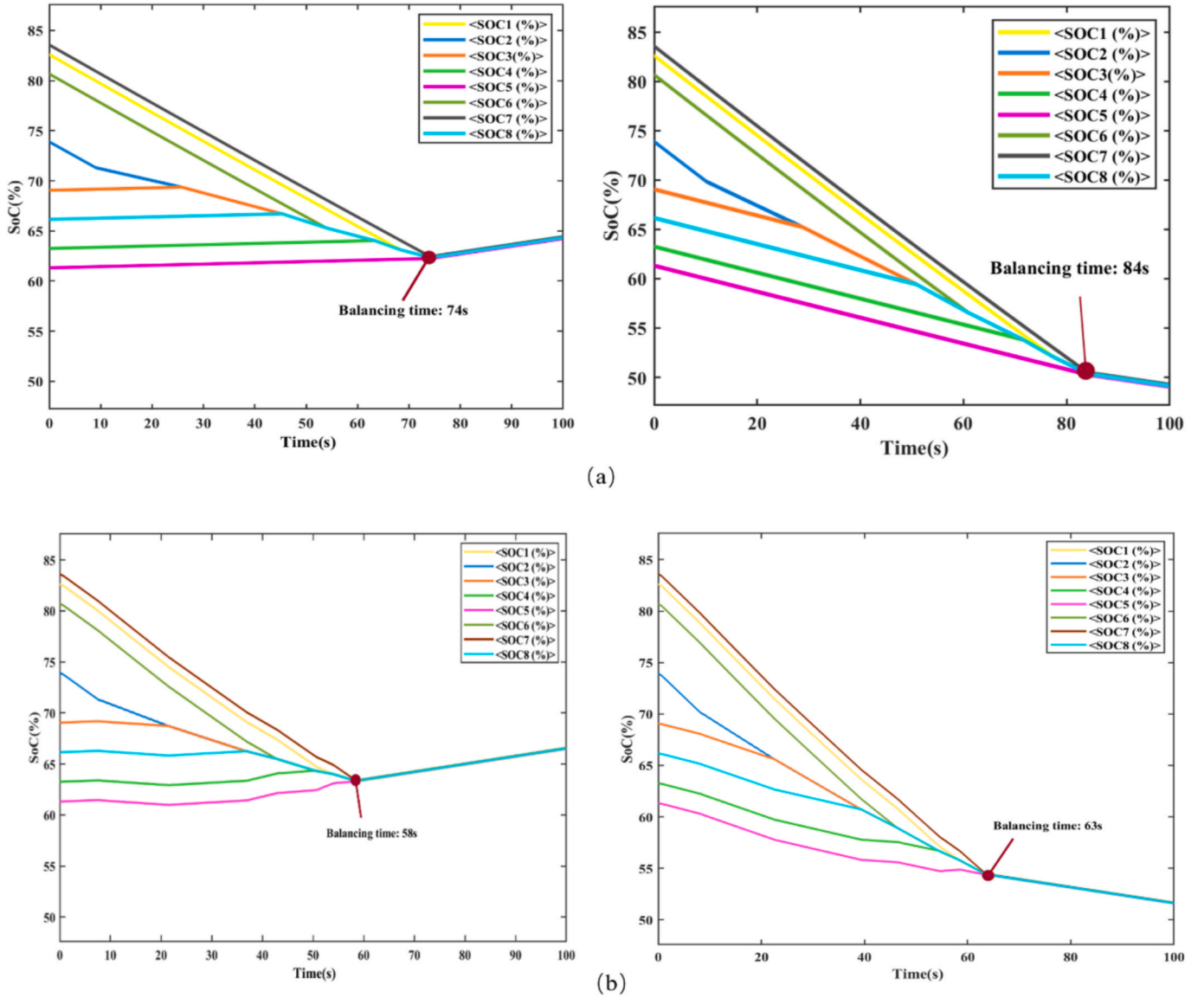


Fig. 19. Simulation results of balancing speed for eight cells under charging and discharging conditions. (a) Buck-boost, (b) Flyback.

Table 5

Balancing time of the four DC-DC converter topologies.

Converter	Category	Balancing time of charging	Balancing time of discharging
Buck-boost	Non-isolated	64 s (4Cells)	73 s (4Cells)
Ćuk	Non-isolated	74 s (8Cells)	84 s (8Cells)
SEPIC	Non-isolated	201 s	201 s
Flyback	Isolated	141 s	142 s
		62 s (4Cells)	67 s (4Cells)
		58 s (8Cells)	63 s (8Cells)

boost converter achieves the fastest balancing speed, whereas the Ćuk converter is the slowest. For isolated flyback operating in a CTPTC structure, the balancing speed is similar to the performance exhibited by the buck-boost converter. These results demonstrate that CTPTC offers higher balancing speeds than CTC configurations, a benefit that becomes more pronounced as the system scales to more cells.

From a ripple perspective, the Ćuk converter's dual-inductor design results in minimal output current ripple but at the cost of slower

balancing. The flyback converter, although generating greater ripple, achieves rapid balancing with the added advantage of isolation. Integrating output filtering in flyback-based equalizers is therefore recommended for future high-performance applications.

4.4. Other comparison

4.4.1. Cost and power loss

Active cell balancing techniques rely on switching elements such as MOSFETs, inductors, capacitors, and transformers, which inevitably introduce switching and conduction losses as well as increase system cost. These factors can significantly influence the overall energy efficiency and economic feasibility of the battery management system, particularly at higher balancing currents or switching frequencies. In the present work, switching losses were not explicitly considered in the simulation models; however, they represent an important aspect in practical applications. To illustrate the cost implications, a four-cell lithium battery pack is considered, with the assumption that the power loss of each identical component type is equivalent. The unit costs of the key components were set according to reported values in the literature [32,73]: MOSFET (M) (0.2), Inductor (L) (0.8), Capacitor (C)

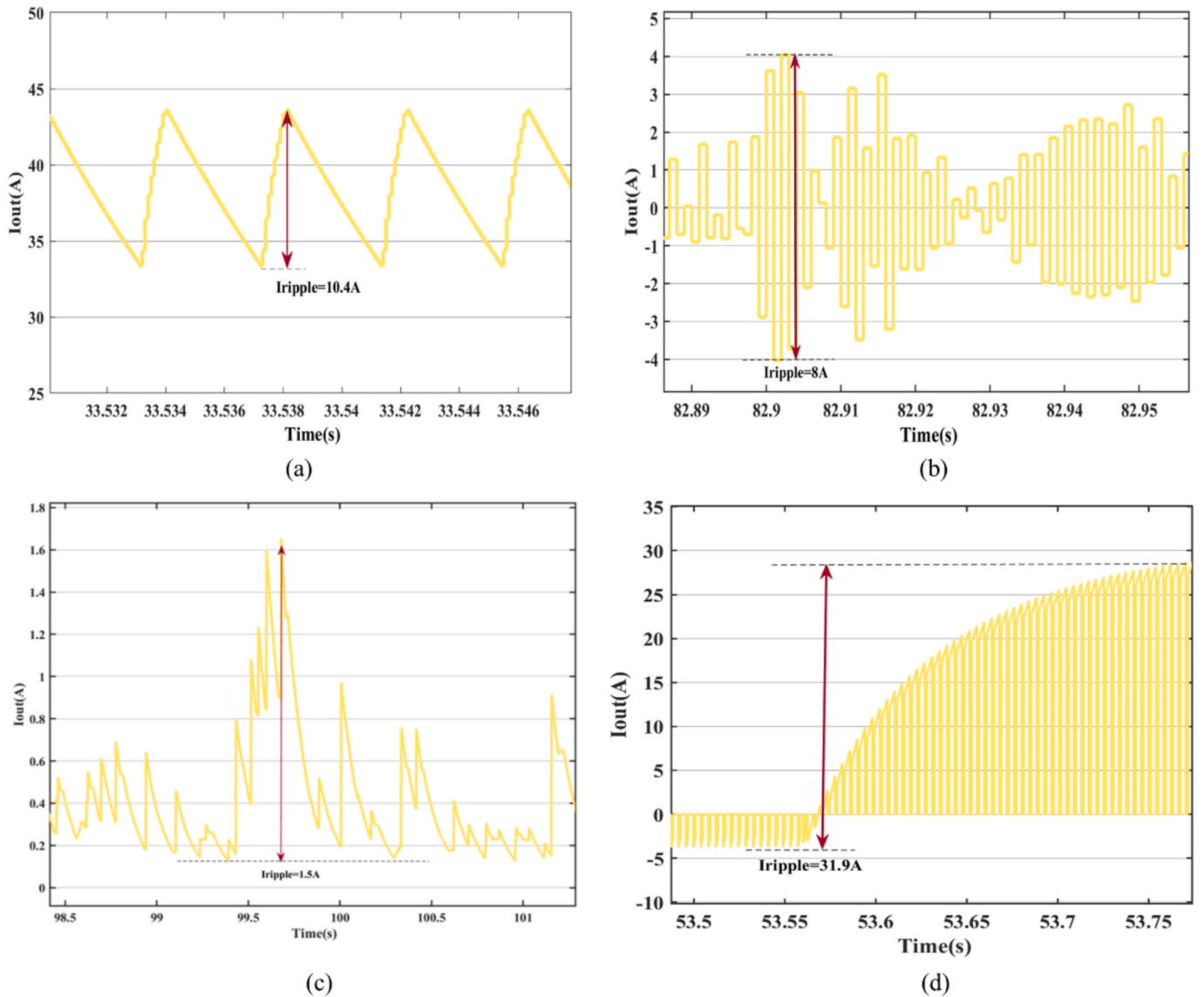


Fig. 20. The current ripple results for the four converter topologies. (a)Buck-boost, (b)SEPIC, (c) Ćuk, (d)Flyback.

Table 6

Evaluation of balancing speed and current ripple for four cells using the SoC-based balancing method.

Converter	Current path	Speed	Current ripple (I_{out})	Battery friendliness
Buck-boost	CTC	fast	medium	medium
Ćuk	CTC	slow	small	excellent
SEPIC	CTC	medium	medium	good
Flyback	CTPTC	fast	large	poor

Table 7

Comparison of estimated component cost and relative power loss for the four balancing topologies.

Converter	Components				Cost (\$)		Power loss
	M	C	L	T			
Buck-boost	8	8	4	0	10.6	Low	Low
Ćuk	8	12	8	0	15.2	High	Medium
SEPIC	8	12	8	0	15.2	High	Medium
Flyback	8	8	0	4	14.4	Medium	High

(0.6), and Single Transformer (T) (2). The specific analysis results are shown in Table 7.

The total power loss is evaluated as the sum of conduction and switching losses of the semiconductor devices, along with copper and core losses in the magnetic components. The buck-boost converter exhibits relatively low losses due to its single-inductor structure, minimal device count, and short energy transfer path, which collectively reduce both conduction and switching dissipation. In contrast, SEPIC and Ćuk converters experience moderate losses: SEPIC primarily because of the extra coupling capacitor and second inductor (or coupled inductor), while Ćuk employs two inductors and an energy-transfer capacitor, leading to increased copper and equivalent series resistance (ESR) losses; the greater number of passive components also contributes to higher circuit cost. The flyback converter shows comparatively high losses in this scenario, mainly resulting from transformer core dissipation, elevated RMS winding currents, and leakage-related energy loss. Nonetheless, at low power levels and switching frequencies, the flyback topology can remain competitive in efficiency, particularly when high-quality magnetic components and optimized rectification schemes are adopted.

4.4.2. Weight-to-power

To evaluate the approximate weight-to-power ratio of the four active balancing topologies, only the major energy-handling components were considered, namely magnetic elements (inductors or transformers), filter/coupling capacitors, and power semiconductors [74]. The rated balancing power for a single cell was taken as:

$$P = V_{\text{nominal}} \times I_{\text{balancing}} = 3.7V \times 2A = 7.4W$$

The component weights were estimated based on typical commercial parts: a 100 μ H power inductor $\approx$ 30 g, a flyback transformer $\approx$ 70 g, a 50 μ F electrolytic capacitor $\approx$ 12 g, a 10 μ F capacitor $\approx$ 2 g, and discrete MOSFET/diode devices for one switching leg $\approx$ 10 g [75,76]. The weight data for inductors, transformers, capacitors, and power semiconductors were derived from commercially available components of comparable ratings using manufacturer datasheets. These values are representative of components with the specified ratings, though variations of approximately $\pm 10\%$ may occur depending on the manufacturer and packaging. The chosen components correspond to typical sizes employed in cell-balancing designs, thereby offering realistic estimates for the weight-to-power ratio analysis. Table 8 summarizes the estimated total mass and corresponding weight-to-power ratio for each topology.

In conventional battery packs that rely solely on passive balancing, the weight of the balancing circuit is negligible, as it consists mainly of resistors and simple switching elements. In contrast, the proposed active balancing circuits—particularly those employing flyback or Ćuk converters—introduce additional weight due to magnetic components and more complex circuitry. While the passive equalizer offers a much lower weight-to-power ratio owing to its minimal component count, it suffers from inherently low balancing efficiency and significant energy dissipation in the form of heat. On the other hand, DC–DC converter–based active equalizers, though heavier, enable bi-directional energy transfer with substantially higher energy utilization efficiency, faster balancing speed, and reduced thermal stress on the cells. Among the active topologies examined, the buck–boost converter exhibits the most favorable weight-to-power ratio, making it an attractive choice for non-isolated applications where both weight and balancing performance are critical.

Based on the above analysis, the comparative results for the four DC–DC topologies are consolidated and presented in the bar chart in Fig. 21, where the ratings 1, 2, and 3 correspond to low/small/slow, medium, and high/large/fast, respectively.

5. Future work

The cell balancing circuit is critical for optimising battery pack performance in terms of capacity uniformity, energy efficiency, and longevity. Continuous innovation in CBC design and enhancement of existing architectures are essential for advancing next-generation energy storage solutions and sustainable electric vehicle technologies. Selecting an appropriate balancing topology requires careful consideration of the specific battery pack characteristics to leverage each design's benefits while mitigating its drawbacks. As higher performance generally entails increased cost, an optimal trade-off between performance metrics and economic constraints must be achieved. The ideal CBC topology maximises balancing efficiency and minimises both power loss and implementation cost.

The design of next-generation CBCs for EVs must address multiple

performance and implementation challenges. To optimize balancing performance, future work should focus on the following key directions:

(1) Selection and optimization of circuit components

Future CBC designs should carefully select switching devices, magnetic components, and capacitors based on the required balancing current, switching frequency, and thermal constraints. Using low-loss and high-temperature-tolerant components can reduce conduction and switching losses. While higher switching speeds enable smaller passive components, they inevitably increase switching losses and current ripple. Thus, proper output filtering (e.g., RC or RCD networks) must be incorporated, especially for bidirectional flyback converters, to suppress ripple without compromising efficiency.

(2) PCB layout and electromagnetic compatibility (EMC)

Proper PCB layout design is critical to minimizing parasitic inductance and resistance, which otherwise distort current ripple and reduce balancing accuracy. High-frequency operation can introduce significant EMI; countermeasures such as optimized grounding, shielded magnetics, and dedicated input/output filtering should be implemented to ensure compliance with automotive EMI standards.

(3) Control strategy enhancements

- **Accurate SoC estimation:** Reliable real-time balancing requires precise state-of-charge estimation. Future control algorithms should integrate advanced SoC estimation methods (e.g., model-based observers or Kalman filters) to improve balancing accuracy, especially under dynamic driving conditions.
- **Temperature-Aware Balancing:** As cell temperature significantly impacts performance and safety, future designs should incorporate temperature monitoring and thermal models into the balancing control logic. Adaptive balancing strategies that consider both SoC deviation and temperature gradients can prevent excessive thermal stress and improve overall pack reliability.
- **Computational Efficiency:** Control algorithms should be designed for periodic sampling with bounded computational load, ensuring that low-power automotive-grade DSPs or micro-controllers can execute balancing tasks in real time without excessive hardware cost.

(4) Topology-level improvements

Current DC–DC converter-based active CBCs face challenges of efficiency and cost. Minimizing passive components and switch count can improve cost-effectiveness, while modular architectures remain desirable for scalability. Among isolated topologies, bidirectional flyback converters remain attractive due to their simplicity and modular control. However, ripple reduction, efficiency optimization, and EMI mitigation are necessary for practical EV deployment.

In summary, future CBC research should focus on achieving an optimal trade-off between balancing efficiency, cost, and implementation feasibility. By integrating accurate SoC estimation, temperature-aware balancing strategies, improved component selection, and robust EMC-aware design practices, the theoretical advantages of active balancing circuits can be effectively translated into reliable, automotive-grade solutions.

Table 8
Comparison of weight-to-power for the four balancing topologies.

Converter	Magnetic elements (g)	Capacitors (g)	Semiconductors (g)	Total mass (g)	Weight-to-power (g/W)
Buck–boost	30	24	20	74	10.00
Ćuk	60	26	20	106	14.32
SEPIC	60	26	20	106	14.32
Flyback	70	24	20	114	15.41

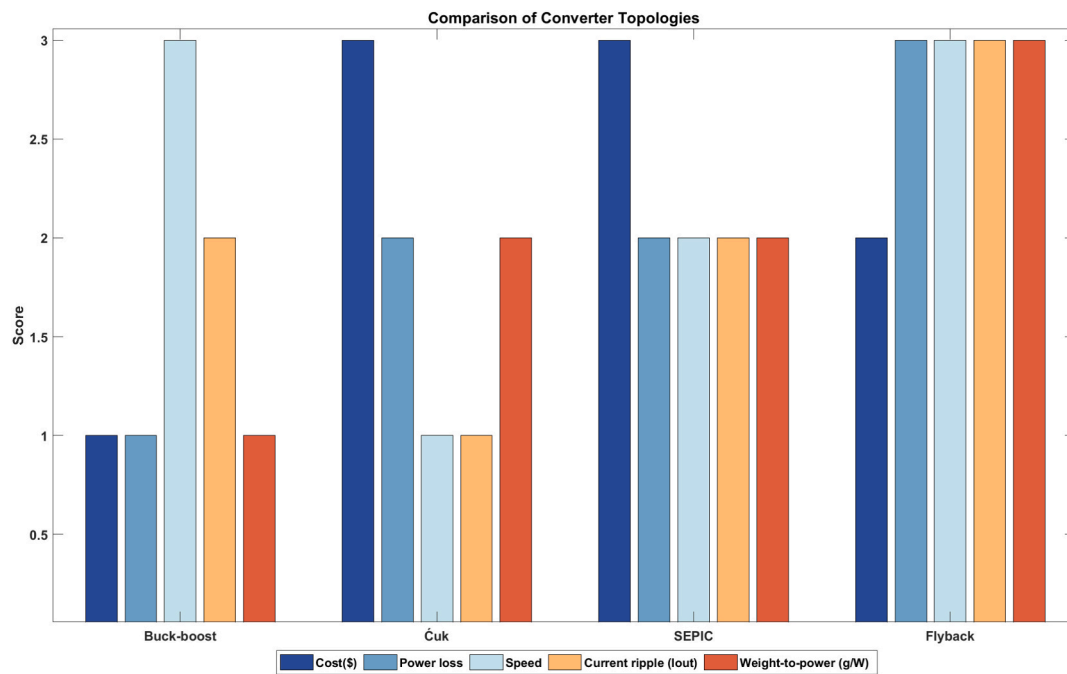


Fig. 21. The comparative results for the four DC–DC topologies.

6. Conclusion

This article presents a comprehensive review of active cell balancing methods, which are critical for optimizing the lifespan and usable capacity of EV battery packs. First, the study introduces a systematic two-fold classification of active balancing techniques, grounded on energy flow direction and the nature of transfer components. Various DC-DC converter-based topologies were subsequently compared with respect to current path, cost, efficiency, control complexity, and balancing time. To further quantify performance, four representative converter topologies were implemented in MATLAB/Simulink and evaluated for balancing speed and output current ripple. The resulting comparative data offer practical guidance for selecting the most appropriate balancing topology in EV applications. In the context of EVs with extensive battery configurations, the CTPTC topology based on DC-DC converters demonstrates balanced performance across multiple metrics, making it a competitive candidate. Although this article provides a systematic survey and simulation-based analysis of current active CBCs, it should be acknowledged that experimental validation has not yet been conducted. This limitation will be addressed in future work, where practical prototypes will be developed and tested to verify the simulation results under real-world operating conditions.

CRediT authorship contribution statement

Hui Li: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Chia Ai Ooi:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization. **Zixiang Jia:** Visualization, Investigation. **Khalid Ammar:** Writing – review & editing. **Mohamad Khairi Ishak:** Writing – review & editing, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data is within the manuscript.

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